

## Revision Notes for Class 7 Mathematics Chapter 8 – Working with Fractions

Fractions are numbers that represent parts of a whole. In this chapter, you will explore different ways to multiply and divide fractions, and learn how to apply these concepts to real-life situations. Understanding how to work with fractions is essential for solving many mathematical problems in daily life, such as measuring quantities or dividing objects equally.

### Multiplication of Fractions

Multiplying fractions involves multiplying the numerators (top numbers) and denominators (bottom numbers) of the two fractions. When multiplying a fraction by a whole number, you can also multiply the numerator by the whole number while keeping the denominator the same. For example, if Aaron's pet tortoise walks  $\frac{1}{4}$  kilometre in one hour, in 3 hours it covers  $3 \times \frac{1}{4} = \frac{3}{4}$  km.

If you multiply a fraction with a whole number, you first divide the whole number by the denominator of the fraction, then multiply the result by the numerator. For example, to find the distance Aaron can walk in  $\frac{2}{5}$  hour at 3 km per hour:  $3 \div 5 = \frac{3}{5}$ , and then multiply by 2 to get  $\frac{6}{5}$  km.

- When multiplying any two fractions, multiply the numerators and denominators:

$$\frac{a}{b} \times \frac{c}{d} = \frac{a \times c}{b \times d}.$$

- Examples:  $\frac{1}{2} \times \frac{1}{4} = \frac{1}{8}$  and  $\frac{3}{4} \times \frac{2}{5} = \frac{6}{20} = \frac{3}{10}$  after simplification.
- For mixed fractions, convert them to improper fractions first before multiplying.

### Multiplying with Lowest Terms and Area Connection

It is often easier to simplify fractions before multiplying. Cancel common factors from numerators and denominators to make calculation simpler. For instance,  $\frac{12}{7} \times \frac{5}{24}$  simplifies to  $\frac{5}{14}$  after cancelling. This process is known as 'apavartana' in Indian mathematics.

Multiplication of two fractions also helps in finding areas. If a rectangle's sides are fractional lengths, such as  $\frac{1}{2}$  unit and  $\frac{1}{4}$  unit, the area becomes the product:  $\frac{1}{2} \times \frac{1}{4} = \frac{1}{8}$  square units.

### Properties of Fraction Multiplication

Multiplication of fractions is commutative, which means the order does not affect the

result:  $\frac{a}{b} \times \frac{c}{d} = \frac{c}{d} \times \frac{a}{b}$ .

- When multiplying two numbers greater than 1, the product is greater than both.
- When one number is a fraction between 0 and 1, the product is less than the whole number.
- Multiplying two fractions less than 1 gives a result smaller than both the fractions.

| Situation                    | Multiplication                   | Relationship                                            |
|------------------------------|----------------------------------|---------------------------------------------------------|
| Both numbers > 1             | $\frac{4}{3} \times 4$           | Product ( $\frac{16}{3}$ ) is greater than both numbers |
| Both numbers between 0 and 1 | $\frac{3}{4} \times \frac{2}{5}$ | Product ( $\frac{3}{10}$ ) is less than both            |

| Situation                      | Multiplication         | Relationship                                                                        |
|--------------------------------|------------------------|-------------------------------------------------------------------------------------|
| One between 0 and 1, one $> 1$ | $\frac{3}{4} \times 5$ | Product ( $\frac{15}{4}$ ) is less than the number $> 1$ but more than the fraction |

So, in summary: when you multiply by a proper fraction, the number gets smaller; when you multiply by something greater than 1, it gets larger.

### Division of Fractions

Dividing one fraction by another involves multiplying by the reciprocal (i.e., flipping numerator and denominator) of the divisor. This is called Brahmagupta's formula:

$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c} = \frac{a \times d}{b \times c}.$$

- To divide by a fraction, invert the divisor and multiply. Example:  

$$\frac{2}{3} \div \frac{3}{5} = \frac{2}{3} \times \frac{5}{3} = \frac{10}{9}.$$
- $1 \div \frac{2}{3} = \frac{3}{2}$  because  $\frac{2}{3} \times \frac{3}{2} = 1$ .
- If dividing by a fraction less than 1, the answer is greater than the starting number. If dividing by a whole number, the answer is smaller.

### Real-Life Applications and Word Problems

Fractions and their operations appear in daily life, such as sharing food, dividing land or money, and calculating time or distance. For example, if  $\frac{1}{4}$  litre of milk is used for 5 cups of tea, each cup gets  $\frac{1}{4} \div 5 = \frac{1}{20}$  litre. If you have 8 m of lace and need to use  $\frac{1}{4}$  m per bag, you can decorate  $8 \div \frac{1}{4} = 32$  bags.

### Example Problems and Practice

- How many glasses of milk in a week if you drink  $1/2$  glass daily?  $7 \times 1/2 = 3.5$  glasses.
- What area will 4 bricks of side  $1/5$  each cover? Area per brick is  $1/5 \times 1/5 = 1/25$  square units.
- If a car travels 16 km per litre of fuel, in  $2\frac{3}{4}$  litres, it will travel  $16 \times 2.75 = 44$  km.

### Reciprocal and Its Use

The reciprocal of a fraction  $\frac{a}{b}$  is  $\frac{b}{a}$ . Multiplying a fraction by its reciprocal always gives 1. Division by a fraction can be solved easily using the reciprocal concept. For example,  $1/6 \div 11/12 = 1/6 \times 12/11 = 2/11$ .

### Historical Notes

Indians were pioneers in systematic operations on fractions, as mentioned by ancient scholars like Brahmagupta and Bhaskaracharya. The method of reducing fractions to lowest terms, called 'apavartana,' and using reciprocals for division are rooted in early Indian mathematics and influenced the world.

### Quick Summary Table

- **Multiplication:**  $\frac{a}{b} \times \frac{c}{d} = \frac{a \times c}{b \times d}$
- **Division:**  $\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c}$
- **Reciprocal:** of  $\frac{a}{b}$  is  $\frac{b}{a}$ .
- When multiplying, cancel common factors before multiplying numerators and denominators.
- If divisor is less than 1, quotient is greater than dividend. If divisor is more than 1, the quotient is less.

### Practice Questions ("Figure it Out")

- **Calculate:**  $7 \times \frac{3}{5}$ ;  $4 \times \frac{1}{3}$ ;  $\frac{9}{7} \times 6$ ;  $\frac{13}{11} \times 6$
- Find area for rectangles with sides like  $3\frac{3}{4}$  ft and  $9\frac{3}{5}$  ft
- **Divide:**  $3 \div \frac{7}{9}$ ;  $\frac{14}{4} \div 2$ ;  $\frac{2}{3} \div \frac{2}{3}$ ;  $\frac{4}{3} \div \frac{3}{4}$

To conclude, mastering fractions is about understanding how to multiply, divide, and simplify them, and applying these skills confidently in daily life and mathematics. The chapter also connects mathematics with ancient Indian contributions, offering engaging puzzles and word problems for meaningful learning.